

Advanced Assignment: Laravel Academic Management & Analytics System

Objective

Design and develop a **role-based academic management system** using **Laravel and MySQL** that supports data management, analytics, and reporting.

In addition to development, students are expected to **think and act in multiple professional roles**, including:

- Software Developer
- Data Analyst
- Data Engineer
- Project Manager

Project Title

Academic Management & Performance Analytics Platform

Technology Requirements

- Laravel (latest version)
- MySQL database
- Blade / Bootstrap (or any frontend framework)
- Chart library (e.g., Chart.js)

Functional Requirements

1. Authentication & Role-Based Access Control

The system must support multiple user roles:

Roles:

- Admin
- Teacher
- Student

Access Control:

- Admin: Full system control
- Teacher: Manage courses and student performance
- Student: View personal academic data

Proper **role-based restrictions** must be enforced across all modules.

2. Student Management Module

Maintain student records with the following details:

- Name
- Email (unique)
- Phone
- Address
- Enrollment Year

Features:

- Create, update, delete, and view students
- Assign students to courses
- Maintain data integrity and uniqueness

3. Course Management Module

Each course should include:

- Course Name
- Course Code (unique)
- Credit Hours
- Assigned Teacher

Features:

- Create and manage courses
- Assign teachers to courses
- Enroll students in courses

4. Performance / Marks Module

Track student academic performance:

Requirements:

- Record marks per student per course
- Automatically assign grades based on marks
- Prevent invalid data entries

5. Database Design

The system must include properly normalized tables such as:

- Users
- Roles
- Students
- Courses
- Enrollments (many-to-many relationship)
- Marks

Relationships must be clearly defined and implemented.

Data Analytics Requirements

Develop a **dashboard** that provides meaningful insights:

Mandatory Analytics:

- Total number of students
- Total number of courses
- Average marks per course
- Pass vs Fail ratio
- Top-performing students

Visualization:

- Use charts (bar, pie, line) to represent data
- Ensure clarity and usability of insights

Data Engineering Requirements

Students must ensure proper data handling:

Data Validation:

- Enforce constraints (e.g., marks between 0–100, unique emails)
- Prevent inconsistent or duplicate data

Data Quality:

- Handle missing or null values
- Maintain referential integrity

Data Handling:

- Use efficient queries
- **Implement pagination for large datasets**
- Optimize database performance where necessary

Project Management Requirements

Students must demonstrate basic project management practices:

Planning:

- Define project scope and objectives
- Identify key features and deliverables

Task Management:

- Break project into phases (design, development, testing, reporting)

Documentation:

Include:

- System overview
- Database design (ER diagram)
- Feature descriptions
- Challenges and solutions

Bonus Features (Optional but Recommended)

- Search and filtering functionality
- Export reports (CSV/PDF)
- REST API endpoints
- Interactive UI enhancements (AJAX)
- Notifications (email or system alerts)

Submission Requirements

Students must submit:

- Complete source code (GitHub or ZIP)
- Database export file (SQL)
- Screenshots of:
 - System modules
 - Role-based access
 - Analytics dashboard
- Project report (3–5 pages) including:
 - System explanation
 - Data insights
 - Project workflow